

Beliz Kaleli

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Sr. Staff Security Research Engineer building AI-powered security systems for large-scale real-world threat detection in high-stakes environments. Primary inventor on 3+ patents, author of 2 award-winning research and 10+ high-impact threat intelligence publications. Proven track record in vulnerability research (CVEs), exploit identification and malicious infrastructure analysis. Core expertise includes designing autonomous AI agents for threat detection, with a focus on emerging attack vectors such as indirect prompt injections, vulnerability scanning, GenAI-driven phishing and malware delivery methods.

Work Experience

Palo Alto Networks	<i>Santa Clara, CA, USA</i>
Sr. Staff Security Research Engineer	Oct 2024 – Present
Staff Security Research Engineer	May 2023 – Oct 2024
Security Researcher Intern	Jun 2022 – Nov 2022

- Led a cross-functional effort to design and develop a hybrid, multi-network vulnerability scanning detection system combining volume-based features, threat signatures, and ML to block ~1M malicious requests daily in real time (patented; first-author paper and blog).
- Built an AI-powered indirect prompt injection (IDPI) detection system protecting AI agents from web-based attacks; achieved first real-world detection of high-severity IDPI (AI reviewer bypass, data exfiltration, prompt leakage, and DoS) (patent pending; invited talk at Stanford University; first-author blog and post).
- Built an autonomous AI agent crawler to solve captchas, authenticate, and navigate complex page interactions to access previously inaccessible content (gated malicious downloads, phishing forms) critical for detection (placed 2nd among 30 teams at the Google Cloud/Palo Alto Networks Agentic AI Hackathon).
- Led a cross-team effort to develop an AI-powered system for detecting OAuth flow abuse in real-time production traffic (patent pending; published 3 posts).
- Designed and implemented an agentic brand impersonation detection system using LLMs with RAG, multimodal analysis, and integrated tools to identify phishing attacks against global and regional brands.
- Primary inventor on 3+ patents (7+ in total) for AI-based security detection techniques, covering phishing, scams, indirect prompt injection, vulnerability scanning, and domain squatting attacks.
- Detected 30+ novel attack campaigns (IDPI, OAuth abuse, browser-in-the-middle, GenAI-based phishing), documented in 9 high-impact threat intelligence publications (6 as first author).

AYESAS	<i>Ankara, Turkey</i>
Software Development Engineer	Sep 2017 – Jun 2018

Skills

Languages: Python, Go, SQL, C, C++, HTML/CSS, JavaScript, Typescript, Java

Tools & Frameworks: GCP, BigQuery, Claude Code, Google ADK, Git, CI/CD, Docker, Kubernetes, REST, gRPC, MongoDB, Scikit-Learn, LangGraph, Windows, Linux, macOS

Awards & Honors

2nd Place at Hackathon <i>Google Cloud/Palo Alto Networks Agentic AI Hackathon</i>	2026
Best Student Paper Award <i>Security and Privacy in Communication Networks (SecureComm)</i>	2024
Best Student Paper Award Finalist <i>The Center for Information & Systems Engineering (CISE)</i>	2020
Best Paper Award <i>Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA)</i>	2019

Patents

• Agentic Web-Based Indirect Prompt Injection (IDPI) Detector	Filed, Apr 2026
<i>B. Kaleli, S. Farooqi, N. Mohamed, O. Starov</i>	
• Detecting Scanning and Attacking URLs in Network Traffic	Patented (US-12483573-B2), Nov 2025
<i>B. Kaleli, F. Liu, O. Starov</i>	
• Phantom Squatting: Anticipating GenAI-Hallucinated URLs Before Attackers Do	Filed, Oct 2025
<i>K. Nagaraj, D. Marty, B. Kaleli, O. Starov</i>	
• Detecting Defensively Registered Domains Using LLMs	Filed, Oct 2025
<i>N. Mohamed, J. Chen, O. Starov, B. Kaleli</i>	

- AVA: Authentication Verification Analyzer for Detecting OAuth Flow Abuses
B. Kaleli, D. Marty, L. Hu, O. Starov
- A Grounded AI-based Pipeline for Detecting Social Engineering Attacks
L. Hu, F. Liu, B. Kaleli, O. Starov
- A Deep Learning Approach for Real-time Scanning URLs Detection Using IUPG Framework
T. Li, B. Kutt, B. Kaleli, F. Liu, P. Peng, O. Starov, W. Hewlett

Filed, Jun 2025

Filed, Jun 2025

Filed, Feb 2025

Academic Papers

- ScanWars: (A Multi-network Approach to Detecting and Analyzing) The Rise of Scanning Activity
B. Kaleli, T. Li, F. Liu, O. Starov, M. Egele, G. Stringhini
Symposium on Electronic Crime Research (eCrime)
- EasyCSPeasy: A Server-side and Language-agnostic XSS Mitigation
B. Kaleli, M. Egele, G. Stringhini
Security and Privacy in Communication Networks (SecureComm) – Best Paper Award
- To Err.Is Human: Characterizing the Threat of Unintended URLs in Social Media
B. Kaleli, B. Kondracki, M. Egele, G. Stringhini, N. Nikiforakis
Network and Distributed System Security Symposium (NDSS)
- Studying the Privacy Issues of the Incorrect Use of the Feature Policy
B. Kaleli, M. Egele, G. Stringhini
Workshop on Measurements, Attacks, and Defenses for the Web (MADWeb)
- On the Perils of Leaking Referrers in Online Collaboration Services
B. Kaleli, M. Egele, G. Stringhini
International Conference on Detection of Intrusions and Malware, and Vulnerability Assessment (DIMVA) – Best Paper Award

2025

2024

2021

2020

2019

Invited Talks & Service

Invited Speaker at Stanford University | Real World AI Security Conference

Program Committee Member | Annual Computer Security Applications Conference (ACSAC)

Invited Speaker at University of Michigan | Center for Computer Security and Society (SECRIT)

2026

2025

2021

Other Publications

Blogs

- Fooling AI Agents: Web-Based Indirect Prompt Injection Observed in the Wild
B. Kaleli, S. Farooqi, O. Starov, N. Mohamed
- Fashionable Phishing Bait: GenAI on the Hook
Z. You, B. Kaleli, L. Hu, W. Wang
- It Was Not Me! Malware-Initiated Vulnerability Scanning Is on the Rise
B. Kaleli, F. Liu, P. Peng, O. Starov, J. Allen, S. Springer

Mar 2026

Aug 2025

Apr 2024

Posts

- New Browser-in-the-Browser Phishing Campaign
B. Kaleli, G. Rane, O. Starov
- Indirect Prompt Injection to Push a Malicious Chrome Extension
B. Kaleli, S. Farooqi, N. Mohamed, O. Starov
- Phishing Campaign Pivots to Sophisticated Methods
B. Kaleli, O. Starov
- Real-World Example of Malicious Indirect Prompt Injection
B. Kaleli, S. Farooqi, O. Starov
- Crypto-themed Emails with Links to Sites Posing as Cryptocurrency Platforms
B. Kaleli, O. Starov
- Phishing Campaigns Impersonate Major Business Platforms and Abuse OAuth Flows
B. Kaleli, O. Starov
- Ongoing Phishing Campaign Using Browser-In-The-Middle Technique
B. Kaleli, O. Starov, L. Hu, S. Shuang, N. Mohamed, B. Duncan

Jun 2026

May 2026

May 2026

Dec 2025

Oct 2025

Oct 2025

Sep 2025

Education

Boston University

Ph.D. & M.S., Computer Engineering, ECE | GPA: 3.8/4.0

Middle East Technical University

B.S., Electrical and Electronics Engineering

Boston, MA, USA

Jan 2025

Ankara, Turkey

Sep 2017